VOLODYMYR POLISHCHUK

Head of Software & Autonomy | Systems Architect | GPS-Denied UAV Navigation

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SUMMARY

Systems architect and engineering leader with 19 years of building production systems at scale, from Wix's developer platform to Playtech's real-time betting engine and UkrCard's payment network. Since 2026 I have led software and autonomy at Applied Technology Collective, a Ukrainian UAV maker, where I designed the GPS-denied navigation its drones fly on, along with the flight-controller firmware and simulator behind it. I reason from first principles, breaking a problem down to its physics, mathematics and computing limits rather than copying existing frameworks. I work AI-native: I own the architecture, the mathematics and the acceptance criteria, and AI coding agents are my implementation team.

EXPERIENCE

Applied Technology Collective | Head of Software & Autonomy

Kyiv, Ukraine | 2026 – Present

Ukrainian developer of strike, loitering-munition and cargo UAVs. I joined as the only software engineer, built the software and autonomy team from scratch and hired its first two engineers. NADIR, SEXTANT and ORRERY are my own IP, used by the company; code walkthrough on request.

NADIR — Visual-Inertial Odometry

- Made the company's drones fly without GPS: visual-inertial odometry turns one downward camera and the autopilot's motion sensors into the position and velocity it would otherwise take from satellites, both for holding position and for guided flight.
- Kept the added hardware to $15 per drone and shipped it as a sealed product: operators calibrate it in the field, and it will not let the drone arm until the setup is safe.

SEXTANT — Long-Range Terrain Navigation

- Extended GPS-denied flight to long range: the drone finds its absolute position and heading by matching the shape of the surrounding terrain against a terrain model, which keeps working in the snow, foliage and changing light that defeat matching by appearance.

ORRERY — Standalone Flight Simulator

- Cut the cost and risk of flight testing with a simulator in which NADIR and SEXTANT must pass their acceptance tests before they reach an aircraft; any ArduPilot version, camera or vehicle is set up by configuration, without code changes.

Wix.com | Principal Software Engineer / Architect

Kyiv, Ukraine | 2017 – 2026

Platformization & Engineering Culture

- Laid the groundwork for opening Wix's internal APIs to third-party developers by leading company-wide adoption of a single API standard, enforced automatically for every service.
- Moved Wix's backend to asynchronous code, cutting its resource costs, through courses and workshops I designed and taught, mandatory for all 1,000+ server engineers.

- Shaped Wix's engineering hiring through 300+ interviews and a vote on its hiring committees.

Platforms & Infrastructure

- Freed about 3,000 developers from managing infrastructure by architecting Wix's internal serverless platform, built with a core team of five; it serves 2,000+ microservices and scales without downtime.
- Brought AI assistance to the Wix App Market by architecting its chat agent.
- Made Wix independent of third-party registrars with its own accredited domain registrar and DNS, automated SSL certificates for 8M+ domains, and led the move of domain management from a legacy monolith to microservices.

Playtech | Software Architect / Tech Lead

Kyiv, Ukraine | 2012 – 2017

- Led a team of five as tech lead, owning the design and technical direction, and moved the sports-betting platform from a monolith to a reactive architecture.
- Kept the platform up through FIFA World Cup peaks of 60M requests per minute while cutting hardware costs, with a design that stays highly available on as few as three servers.
- Met regulation that ruled out the cloud by designing a distributed data store that runs on the company's own servers and serves reads locally.

UkrCard | Systems Architect

Kyiv, Ukraine | 2007 – 2012

- Secured the payment network by leading the move of every ATM, POS and self-service terminal from legacy wired lines to encrypted wireless connections.
- Cut costly false positives in fraud detection with a neural-network assistant that learned from operator feedback (Master's project).

SKILLS

Autonomy: visual-inertial navigation, sensor fusion, terrain-referenced navigation, ArduPilot, MAVLink.
Engineering: Rust, C, Scala, Java, Python; embedded Linux, distributed systems, Kubernetes, gRPC.
AI: RAG, LLM applications, neural networks from first principles.

EDUCATION

M.Sc. in Cybernetics | Taras Shevchenko National University of Kyiv, Faculty of Cybernetics. Thesis: Ensuring the Dependability of Transaction Classification in a Payment System.

B.Sc. in Applied Mathematics | Taras Shevchenko National University of Kyiv, Faculty of Cybernetics

MILITARY & OTHER QUALIFICATIONS

Armed Forces of Ukraine (reserve): Junior Lieutenant, military specialty 708200: operation, tactical deployment and maintenance of air-defense radar systems.
IYT Skipper (International Yacht Training): practical navigation.
Interests: world models for robot navigation (V-JEPA, Genesis); basal cognition (Michael Levin); foundations of physics.